

Gestational Diabetes Mellitus

Diabetes in pregnancy

Diabetes in pregnancy occur in the following situations;

- When a woman with type 1 or type 2 diabetes mellitus gets pregnant
- When a pregnant woman develops gestational diabetes

Gestational diabetes mellitus

Gestational diabetes is a type of diabetes that occurs during the second or third trimester of pregnancy. That is between month four and month nine of pregnancy.

Risk factors of gestational diabetes

The following factors put women at greater risk of developing gestational diabetes;

- Being 35 years and above
- Being overweight or obese before pregnancy
- Gaining a lot of weight during pregnancy
- Having pre-diabetes before pregnancy
- Had a baby who was more than 4kg (8.8 pounds) at birth
- A parent, sister, or brother with type 2 diabetes
- Had polycystic ovarian syndrome (PCOS) before pregnancy
- Used corticosteroids consistently and continuously for a long time

These are factors that also increase the risk of developing type 2 diabetes mellitus before pregnancy.

How to know that you have gestational diabetes (diagnosing gestational diabetes)

If a pregnant woman has one of the risk factors for gestational diabetes before pregnancy. It is helpful to monitor her blood sugar regularly to determine whether it is increasing above normal levels. This is because it puts her life and her baby at risk.

The steps of diagnosis of gestational diabetes are done through three steps

Step one is to screen with a fasting blood sugar. If it is above 5.3 mmol/L (95 mg/dL), then it signals that the blood sugar is increasing, and there is a need to confirm if it is gestational diabetes.

The second step is to conduct an initial 50 g glucose challenge test; if it is abnormal. An abnormal result is having a blood sugar of more than or equal to 10.6mmol/L after one hour and then more than or equal to 9.0mmol/L after two hours.

If any of these results are abnormal, then the third step is a 75g oral glucose tolerance test is done. If any of the results are abnormal, then a diagnosis of gestational diabetes mellitus (GDM) is confirmed.

Another approach is to use a 1-step method with a 75 g oral glucose tolerance test. A diagnosis of GDM is made if one plasma glucose value is abnormal results (having a blood sugar of more than or equal to 10.0 mmol/L after one hour and then more than or equal to 8.5 mmol/L after two hours).

How do you treat gestational diabetes

Gestational Diabetes is treated based on the blood sugar level at the time when it was found out or diagnosed. The treatment is done in three steps;

Step one: First, physical activity and diet will be recommended to bring down the blood sugar level to normal level. Then the fasting blood sugar is monitored to see if this works. If it works, it is continued. But if it does not work, then medications that lower blood sugar will be recommended.

Step two: Second, if physical and diet are not able to bring down the blood sugar to normal level, then a medication that lowers blood sugar is recommended. Only two drugs are considered safe for both the baby and the pregnant woman. It is either metformin or insulin that will be recommended. This will be done by a doctor and the clinical care team. The blood sugar is monitored regularly to see if this is working and the blood sugar is coming down to normal level. If it does not work, then another medication is needed.

Step three: If one drug which is either metformin or insulin does not work, then another medication is added. If insulin was used at first, then metformin is added. If metformin was used at first, then insulin is added.

What kind of food is recommended for gestational diabetes

A balanced, healthy diet is recommended to manage gestational diabetes.

Eat about 5-6 servings of non-starchy vegetables every day.

Eat more of whole grains, unsaturated fat, legumes, and lentils.

Avoid sugar, limit saturated fat or trans fats and reduce salt intake. Avoid processed carbs like bread and other pastries.

Avoid alcohol and smoking during pregnancy.

What kind of exercise is recommended for gestational diabetes

Exercise is important for reducing blood sugar. It is important to combine aerobic exercise and strength training exercises during pregnancy. The intensity of exercise depends on your health status. It is important to discuss with your doctor before you start exercising.

About 150 minutes of moderate-intensity exercise is recommended a week. It is also important to include muscle-strengthening and stretching exercises to improve or maintain strength and flexibility during pregnancy.

How does a pregnant woman with gestational diabetes manage weight during pregnancy

Gaining some weight during pregnancy is very important. The amount of weight a pregnant woman gains depends on her weight before pregnancy. The table below shows the normal weight gain to expect. If you gain too much weight during pregnancy, discuss it with your doctor.

	Normal weight gain per week		Total weight gain	
	Kg/week	lb/week	Kg	lbs
BMI less than 18.5	0.5	1.0	12.5 - 18	28 - 40
BMI 18.5 to 24.9	0.4	1.0	11.6 - 16	25 - 35
BMI 25 to 29.9	0.3	0.6	7 – 11.5	15 - 25
BMI more than 30	0.2	0.5	5 - 9	11 -20

How does a pregnant woman with gestational diabetes monitor her blood sugar

Monitoring blood sugar regularly is very important. This helps you to know when it is too high and when it is too low. Checking at home with a glucometer is recommended. If you don't have a glucometer, you should monitor whenever you go for ante-natal care.

Monitoring before and after meals can help you recognize which foods increase your blood sugar and avoid them.

If you are using insulin, you should also consider monitoring more often.

HBA1C is not routinely used to monitor blood sugar control in gestational diabetes.

How should a pregnant woman with gestational diabetes monitor the baby

It is important to monitor how well the baby is developing. This can be done with ultrasound scan. It is recommended to have ultrasound scans before 24 weeks to find out how far along the pregnancy is. The ultrasound scan will help to detect if the baby has any abnormalities.

You also need to monitor the movements the baby is making to know when it normal movements or when it reduces or increases. It is important to notify your doctor once you notice abnormal movements.

You may need to use ultrasound scan to monitor more often as you approach your expected delivery date.

Ante-natal care advice for gestational diabetes

It is important to start ante-natal care as soon as you know that you are pregnant. Women with gestational diabetes will require additional support to eat well, exercise regularly, manage weight, reduce blood sugar, and monitor their health and the baby.

How should a pregnant woman with gestational diabetes prepare for delivery.

Follow the guidance of the doctor, midwife and clinical care team to prepare for delivery. At the beginning of ante-natal care, they will calculate your expected delivery date

(EDD). Take note of your EDD. As the EDD approaches, the doctor and the care team, will provide additional guidance.